

Data Exploration: HCC Tumor vs Matched Normal Liver

Team

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Dataset

RNA-seq of 50 Paired Hepatocellular Carcinoma Samples

Source: <https://www.refine.bio/experiments/SRP068976/rna-seq-of-50-paired-hepatocellular-carcinoma?ref=search>

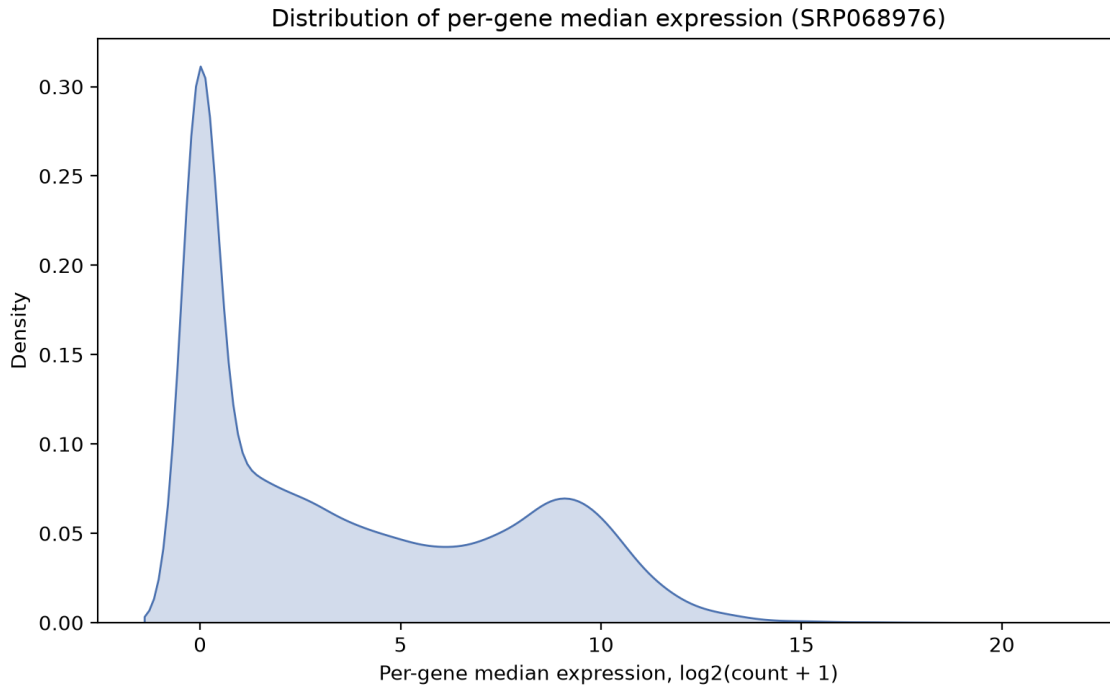
Scientific Question

Can gene expression profiles distinguish hepatocellular carcinoma (HCC) tumor tissue from matched non-tumor liver tissue?

GitHub Repository

<https://github.com/JunhaoLiXD/HCC-RNASeq-Classification>

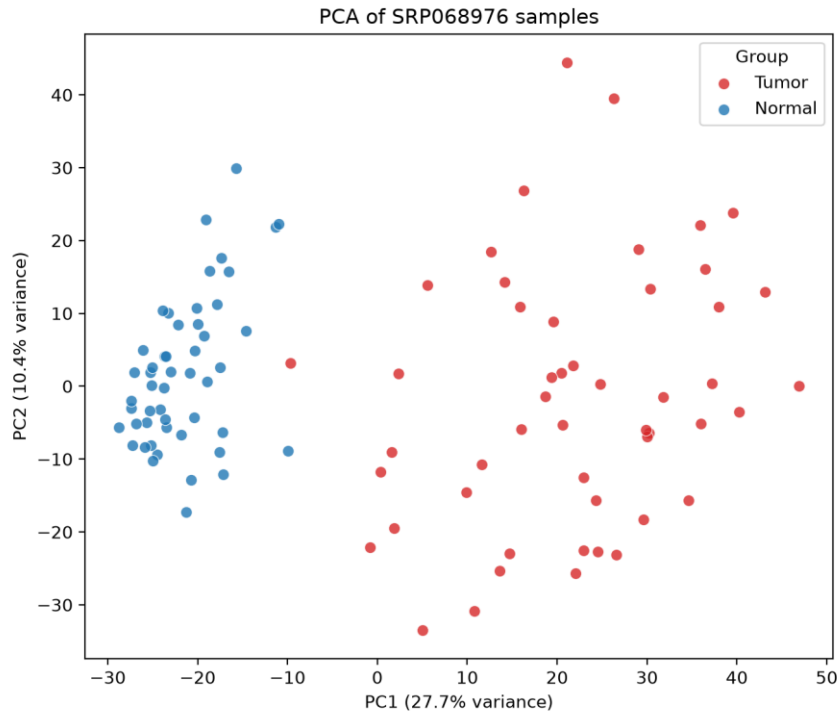
1. Data loading, gene annotation, and per-gene variation



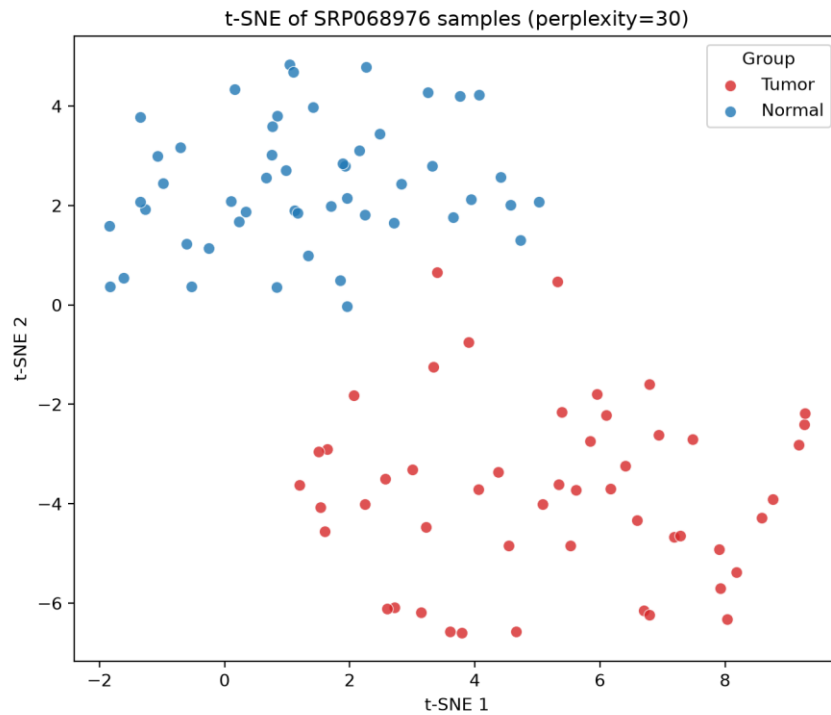
We loaded the SRP068976 expression matrix (43,355 Ensembl genes x 100 samples, 50 tumor and 50 matched normal) and mapped Ensembl IDs to gene symbols with mygene, annotating 33,537 genes (77.4%). After a $\log_2(\text{count} + 1)$ transform we computed each gene's median expression across samples and plotted its density. The distribution is clearly bimodal: a large peak at $\log_2 \sim 0$ corresponds to ~15,000 lowly-expressed or silent genes, while a second broad peak near $\log_2 \sim 9$ represents the actively expressed transcriptome. This wide dynamic range confirms substantial variation and motivates filtering low-count genes downstream.

2. Dimensionality reduction (PCA, t-SNE, UMAP)

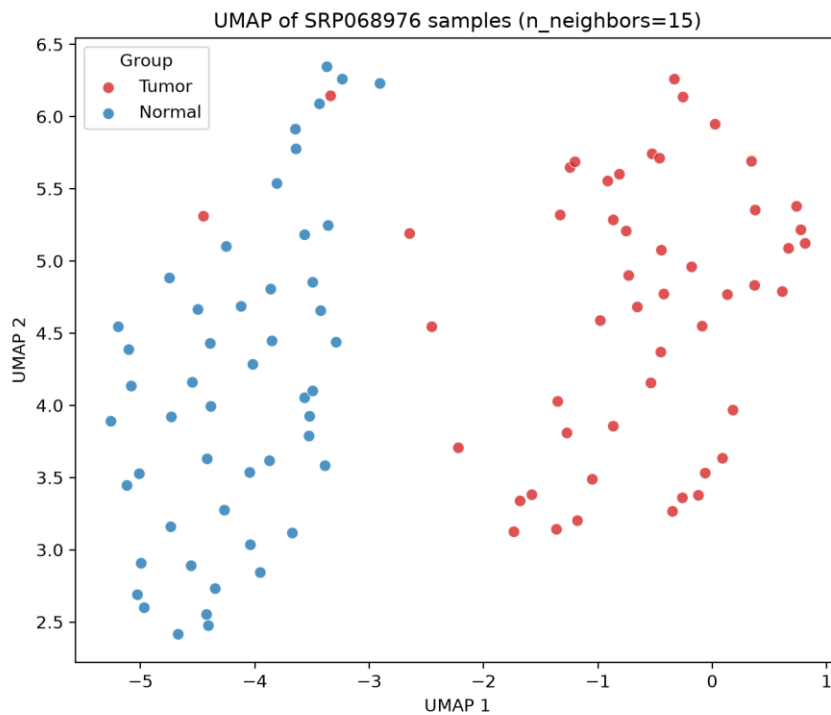
For all three methods we filtered to genes with more than 10 counts in at least 10 samples (23,393 genes), log2-transformed them, kept the 2,000 most-variable genes, and z-scored each gene before projecting the 100 samples into two dimensions.



PCA. This linear projection separates the groups almost entirely along PC1, which explains 27.7% of the variance, while PC2 explains only 10.4% and reflects within-group variation. Tumor samples fall on the right and normal on the left, showing that tumor-vs-normal status is the single largest source of variation in the data.

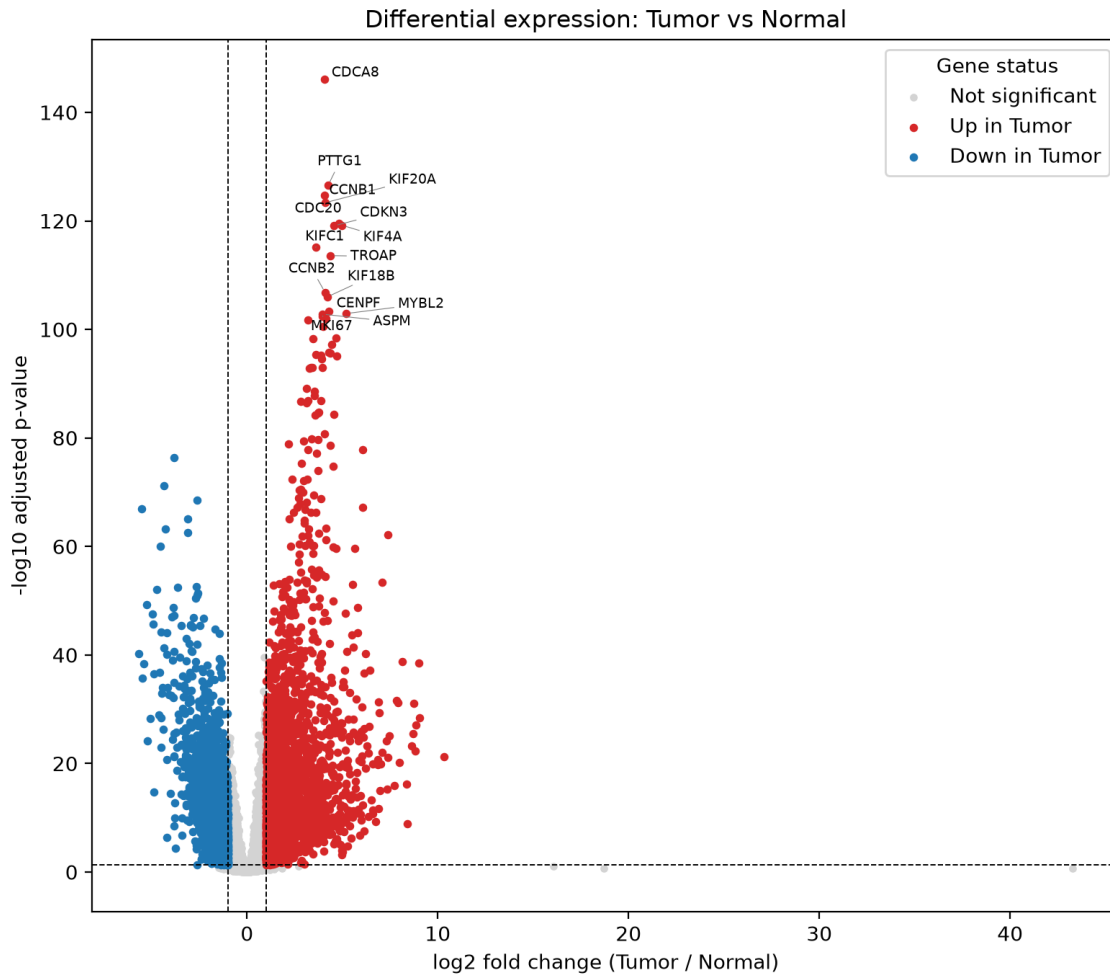


t-SNE. Using perplexity 30, t-SNE models non-linear local structure and produces two tight, well-separated clusters - normal samples group at the top and tumor samples at the bottom - with only one or two tumor samples falling near the normal cluster.



UMAP. With `n_neighbors` 15, UMAP likewise yields two clearly separated groups, normal on one side and tumor on the other. Like t-SNE it gives tighter, more visually distinct clusters than the linear PCA, confirming the two tissue types have systematically different expression profiles.

3. Differential expression (Tumor vs Normal, PyDESeq2)

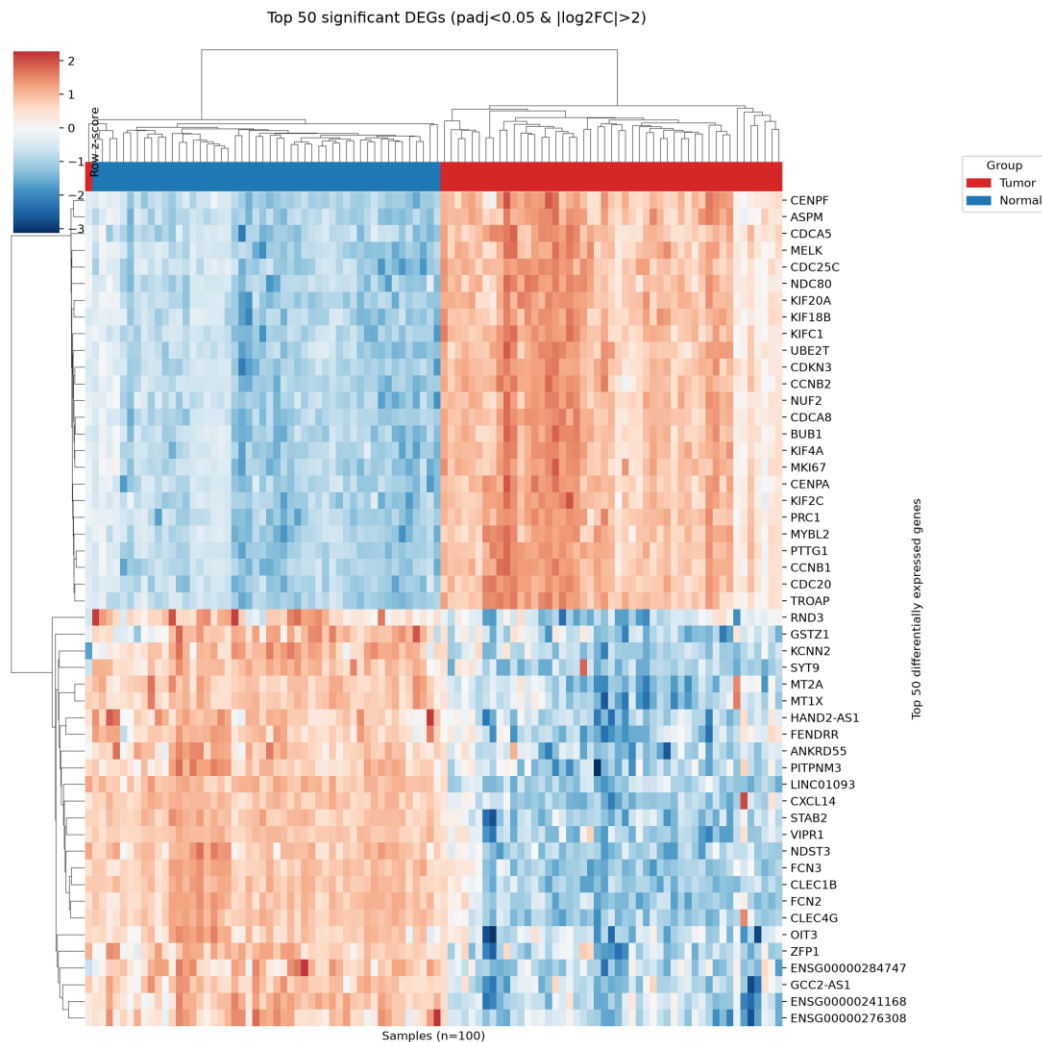


We tested differential expression with PyDESeq2 using a two-group design (`~condition`, reference = Normal); 23,393 genes passing the low-count filter were rounded to integer counts as DESeq2 requires. At $\text{padj} < 0.05$, 15,202 genes were significant (3,607 up and 2,212 down in tumor at $|\log_2\text{FC}| \geq 1$). The most significant genes are cell-cycle and proliferation markers up-regulated in tumor, including CDCA8, PTTG1, CCNB1, CDC20, KIF20A, and MYBL2, aligning well with the established molecular hallmarks of HCC.

Top 50 differentially expressed genes (by adjusted p-value):

gene_symbol	baseMean	log2FoldChange	pvalue	padj
CDCA8	1.687e+02	4.075e+00	3.065e-151	7.169e-147
PTTG1	2.729e+02	4.260e+00	2.194e-131	2.566e-127
CCNB1	3.973e+02	4.064e+00	2.552e-129	1.990e-125
KIF20A	2.011e+02	4.104e+00	7.581e-128	4.433e-124
CDC20	4.027e+02	4.827e+00	6.342e-124	2.967e-120
KIF4A	2.137e+02	4.980e+00	1.665e-123	6.492e-120
CDKN3	1.652e+02	4.556e+00	2.461e-123	8.226e-120
KIFC1	2.757e+02	3.596e+00	2.545e-119	7.441e-116
TROAP	3.724e+02	4.374e+00	9.597e-118	2.494e-114
CCNB2	2.031e+02	4.094e+00	6.531e-111	1.528e-107
KIF18B	2.572e+02	4.211e+00	4.734e-110	1.007e-106
CENPF	5.590e+02	4.283e+00	2.758e-107	5.376e-104
MYBL2	4.762e+02	5.215e+00	6.520e-107	1.173e-103
ASPM	5.144e+02	3.939e+00	9.050e-107	1.512e-103
MKI67	7.104e+02	3.961e+00	2.186e-106	3.409e-103
CDCA5	2.108e+02	4.135e+00	7.536e-106	1.102e-102
NDC80	1.736e+02	3.190e+00	1.238e-105	1.704e-102
MELK	1.304e+02	3.989e+00	2.382e-104	3.095e-101
CENPA	7.166e+01	4.653e+00	2.957e-102	3.640e-99
PRC1	3.175e+02	3.456e+00	3.881e-102	4.540e-99
CDC25C	7.601e+01	4.430e+00	4.984e-101	5.552e-98
KIF2C	2.238e+02	4.273e+00	1.704e-99	1.812e-96
NUF2	1.705e+02	4.359e+00	2.199e-99	2.236e-96
UBE2T	1.648e+02	3.622e+00	4.382e-99	4.271e-96
BUB1	1.687e+02	3.861e+00	7.040e-99	6.588e-96
TOP2A	1.220e+03	4.704e+00	9.970e-99	8.971e-96
CDK1	2.519e+02	3.919e+00	3.339e-98	2.893e-95
TPX2	5.902e+02	3.348e+00	1.114e-96	9.205e-94
RRM2	4.348e+02	3.410e+00	1.141e-96	9.205e-94
NEK2	1.490e+02	3.949e+00	1.542e-96	1.202e-93
GTSE1	1.500e+02	3.290e+00	2.030e-96	1.532e-93
CENPW	1.206e+02	3.138e+00	1.050e-92	7.679e-90
SPC24	1.662e+02	3.528e+00	3.126e-92	2.216e-89
ASF1B	1.427e+02	3.527e+00	2.233e-91	1.536e-88
NCAPG	3.018e+02	3.883e+00	2.019e-90	1.349e-87
ZWINT	3.460e+02	3.186e+00	2.207e-90	1.434e-87
RACGAP1	3.266e+02	2.828e+00	3.138e-90	1.984e-87
TCF19	3.946e+02	3.127e+00	6.398e-90	3.939e-87
BIRC5	4.238e+02	3.767e+00	2.820e-88	1.692e-85
ANLN	2.103e+02	3.707e+00	3.766e-88	2.202e-85
ESM1	1.094e+02	4.571e+00	8.592e-88	4.902e-85
NCAPH	1.083e+02	3.581e+00	1.188e-87	6.615e-85
E2F1	3.793e+02	4.046e+00	2.885e-84	1.569e-81
HJURP	2.341e+02	3.400e+00	2.564e-83	1.363e-80
AURKB	2.019e+02	3.707e+00	4.116e-83	2.140e-80
ECT2	2.347e+02	2.957e+00	7.719e-83	3.926e-80
RFC4	3.604e+02	2.172e+00	2.388e-82	1.188e-79
DLGAP5	1.153e+02	4.361e+00	5.517e-82	2.689e-79
GPC3	8.485e+03	6.072e+00	3.484e-81	1.663e-78
CDCA3	2.467e+02	3.208e+00	3.589e-81	1.679e-78

4. Heatmap of significant DEGs



We defined a strict DEG set as $\text{padj} < 0.05$ AND $|\log_2\text{FC}| > 2$ (1,678 genes; full list in the repository). For readability the heatmap shows the 50 most significant of these (top 25 up and top 25 down), z-scored per gene, with a top side bar colored by group. Unsupervised column clustering splits the 100 samples almost perfectly into a normal block and a tumor block. Up-regulated genes are proliferation markers (CENPF, ASPM, MKI67) while down-regulated genes include normal liver-function genes (GSTZ1, CXCL14, FCN2, CLEC4G), consistent with loss of hepatocyte identity.

5. Gene set enrichment analysis (two methods)

As a team of one, two methods from the instructor's list were run, both on the same ontology (Gene Ontology biological process) so the combined table is meaningful. Method 1: g:Profiler (gProfiler2) over-representation of the 1,678 strict DEGs against GO:BP (g:SCS correction) yielded 247 significant terms, led by mitotic cell cycle,

nuclear division and chromosome segregation. Method 2: a Wilcoxon rank-sum (Mann-Whitney U) gene-set test ranked all genes by the DESeq2 Wald statistic and tested each GO:BP set (sizes 10-500; BH correction), giving 450 significant terms led by translation, ribosome biogenesis, DNA repair and mitotic chromosome segregation. Full tables are in results/tables/.

6. Combined enrichment table

The two GO:BP results were merged on GO term id into a joint table of 3,532 unique terms (full table in the repository). Each row holds each method's adjusted p-value, the number of methods that tested the term, and the number that called it significant. 64 terms are significant in both methods and 569 in exactly one. Note g:Profiler only reports significant terms, so its presence is treated as tested-and-significant; the Wilcoxon method reports every tested term.

7. Top 10 shared enriched terms

The top 10 terms enriched in both methods are all mitosis / cell-cycle processes. Two methodologically different approaches converge on the same biology, giving high confidence that the dominant difference between HCC tumor and matched normal liver is activation of the mitotic cell-division program.

term_id	term_name	gprofiler_padj	wilcoxon_padj	n_methods_tested	n_methods_significant
GO:0000070	Mitotic Sister Chromatid Segregation	7.824e-17	2.388e-19	2	2
GO:0140014	Mitotic Nuclear Division	2.154e-17	8.296e-10	2	2
GO:0000819	Sister Chromatid Segregation	1.345e-16	1.222e-09	2	2
GO:0098813	Nuclear Chromosome Segregation	1.508e-19	2.400e-04	2	2
GO:0010564	Regulation of Cell Cycle Process	2.429e-11	1.972e-09	2	2
GO:0051310	Metaphase Chromosome Alignment	7.727e-09	2.223e-11	2	2
GO:0007094	Mitotic Spindle Assembly	9.045e-11	9.448e-09	2	2

	Checkpoint Signaling				
GO:0071173	Spindle Assembly Checkpoint Signaling	9.045e-11	9.448e-09	2	2
GO:0071174	Mitotic Spindle Checkpoint Signaling	9.045e-11	9.448e-09	2	2
GO:0007052	Mitotic Spindle Organization	1.069e-06	3.460e-12	2	2